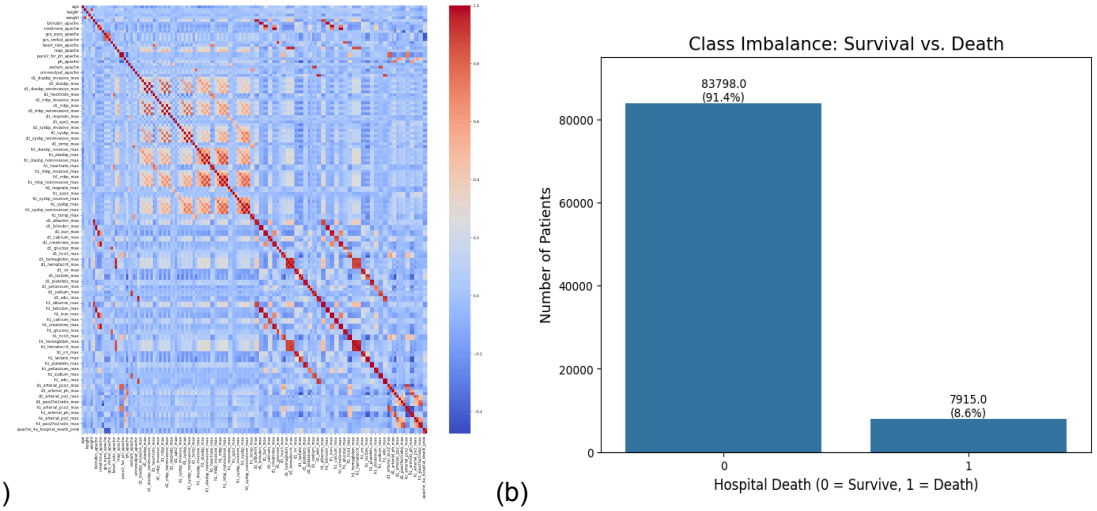
Supplementary Figures

	A	B	C	D	E	F	G	H	I	J	K	L
1	hospital_death	encounter_id	patient_id	hospital_id	age	bmi	elective_surgery	ethnicity	gender	height	hospital_admit_source	icu_admit_source
2	0	66154	25312	118	68	22.73	0	Caucasian	M	180.3	Floor	Floor
3	0	114252	59342	81	77	27.42	0	Caucasian	F	160	Floor	Floor
4	0	119783	50777	118	25	31.95	0	Caucasian	F	172.7	Emergency Department	Accident & Emergency
5	0	79267	46918	118	81	22.64	1	Caucasian	F	165.1	Operating Room	Operating Room / Recovery
6	0	92056	34377	33	19	NA	0	Caucasian	M	188		Accident & Emergency
7	0	33181	74489	83	67	27.56	0	Caucasian	M	190.5	Direct Admit	Accident & Emergency
8	0	82208	49526	83	59	57.45	0	Caucasian	F	165.1	Operating Room	Accident & Emergency
9	0	120995	50129	33	70	NA	0	Caucasian	M	165	Emergency Department	Accident & Emergency
10	1	80471	10577	118	45	NA	0	Caucasian	M	170.2	Other Hospital	Other Hospital
11	0	42871	90749	118	50	25.71	0		M	175.3	Direct Admit	Accident & Emergency
12	0	105427	125898	77	72	28.257052	1	Hispanic	F	154.9	Operating Room	Operating Room / Recovery
13	0	91609	78266	83	80	27.382813	1	Caucasian	F	160	Operating Room	Operating Room / Recovery
14	0	76356	41311	118	48	NA	0	Caucasian	M	175.3	Emergency Department	Accident & Emergency
15	0	34030	103766	118	65	NA	1	Caucasian	M	172.7	Operating Room	Operating Room / Recovery
16	0	108388	98174	118	81	38.189067	1	Caucasian	M	172.7	Operating Room	Operating Room / Recovery
17	0	120677	124688	118	78	NA	0	Caucasian	F	160	Emergency Department	Accident & Emergency
18	0	115771	71252	81	30	23.383178	0	Caucasian	M	193	Direct Admit	Accident & Emergency
19	0	22471	112115	118	46	25.845717	0	Hispanic	M	167.6	Emergency Department	Accident & Emergency
20	0	48056	114220	118	65	28.408929	0	Hispanic	M	167.6	Emergency Department	Accident & Emergency
21	0	45954	102946	118	39	37.010278	1		M	177.8	Operating Room	Operating Room / Recovery
22	0	79115	88295	83	79	28.3761	1	Caucasian	F	170.2	Operating Room	Operating Room / Recovery
23	0	118620	59115	118	80	35.659989	0	Caucasian	M	170.2	Floor	Floor
24	0	21050	31257	81	58	NA	1	Caucasian	F	162.6		Operating Room / Recovery
25	0	95460	120539	118	87	21.963763	0	Caucasian	M	180.3	Floor	Floor
26	0	7220	92453	77	60	29.509959	0	Caucasian	M	188	Direct Admit	Accident & Emergency
27	1	29208	114628	118	68	26.010703	0	Caucasian	F	165.1		Floor
28	0	32902	17922	118	85	23.80977	0	Caucasian	F	152.4	Emergency Department	Accident & Emergency

Fig. 1: Snapshot of 'training_v2.csv' (target variable 'hospital_death' highlighted in yellow). Despite the file name, this is our total data set, with over 90k entries



(a) (b)

Fig. 2: (a) Heat map of multicollinearity in the unrefined/unselected predictor set. (b) Class imbalance of our target variable, with a large majority of patients surviving their hospital stay.

```
--- Lambda Index: 4, C=0.0004, Actual Lambda=2223.00 ---
21 non-zero coefficients:
  solid_tumor_with_metastasis_0.0: -0.511082
  gcs_unable_apache_0.0: -0.488972
  ventilated_apache_0.0: -0.437755
  cirrhosis_0.0: -0.231704
  leukemia_0.0: -0.220356
  gcs_motor_apache: -0.212406
  immunosuppression_0.0: -0.209895
  hepatic_failure_0.0: -0.187396
  d1_bun_min: 0.138070
  intubated_apache_0.0: -0.135993
  d1_spo2_min: -0.135919
  d1_sysbp_min: -0.109696
  d1_temp_min: -0.086598
  d1_heartrate_max: 0.045690
  h1_resprate_min: 0.038139
  d1_mbp_min: -0.031029
  gcs_eyes_apache: -0.016665
  d1_sysbp_noninvasive_min: -0.013537
  d1_hco3_min: -0.009136
  age: 0.007597
  d1_mbp_noninvasive_min: -0.007065
```

Fig. 3: List of 21 features selected by Lasso-regularized logistic regression

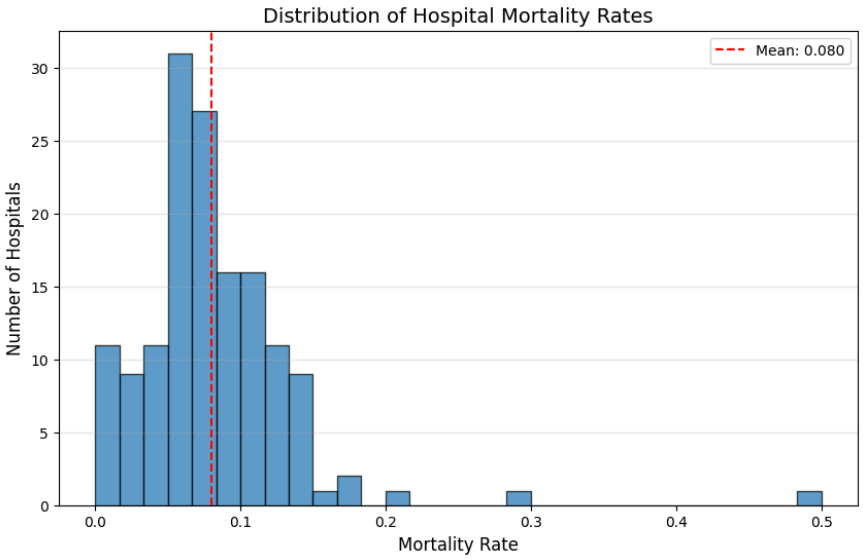


Fig. 4: Distribution of hospital mortality, by hospital

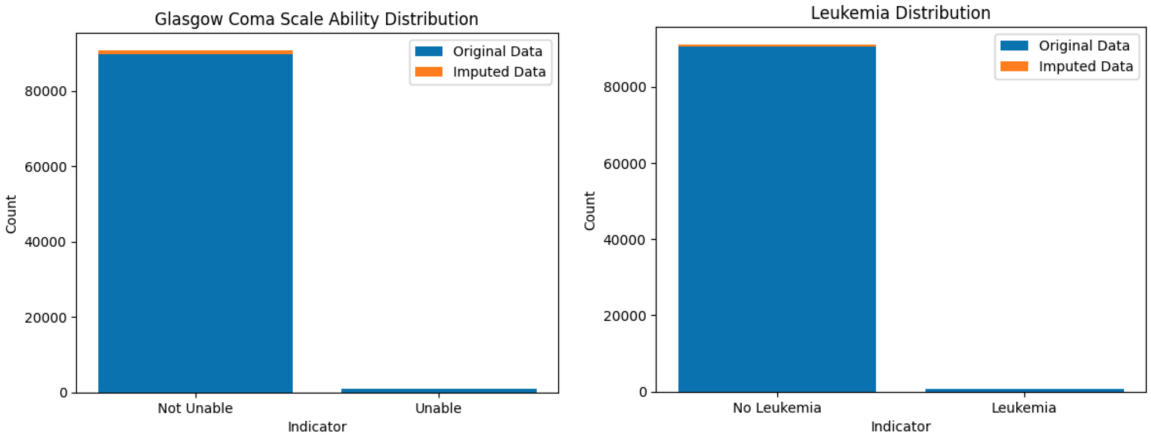


Fig. 5: Example distributions of binary predictor variables. Almost all Lasso-selected predictor variables are imbalanced as such.

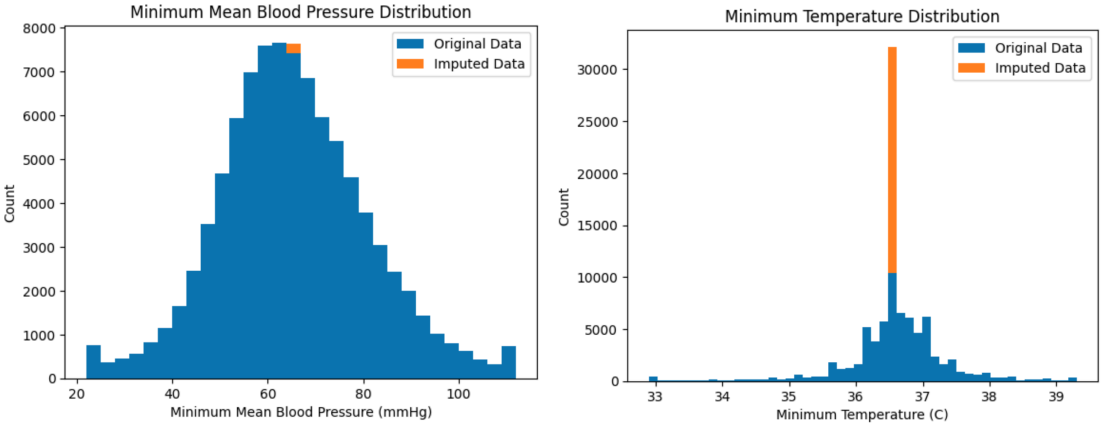


Fig. 6: Example distributions of Lasso-selected numeric predictor variables.

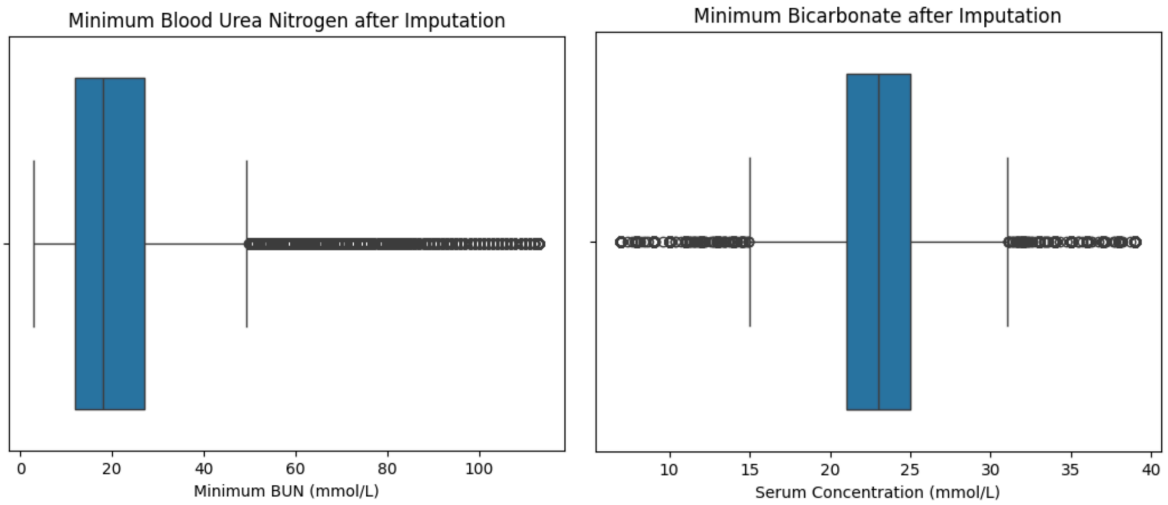


Fig. 7: Example box plots of distributions of Lasso-selected numeric predictor variables.

	feature	VIF	feature	VIF	feature	VIF
0	d1_sysbp_noninvasive_min	95.805603	intubated_apache_0.0	1.649647	d1_hco3_min	1.122899
1	d1_sysbp_min	94.607378	hepatic_failure_0.0	1.385902	immunosuppression_0.0	1.104354
2	d1_mbp_noninvasive_min	76.527622	cirrhosis_0.0	1.385572	d1_spo2_min	1.094054
3	d1_mbp_min	75.186678	d1_heartrate_max	1.198955	solid_tumor_with_metastasis_0.0	1.082204
4	gcs_eyes_apache	3.018682	age	1.142376	h1_resprate_min	1.060167
5	gcs_motor_apache	2.877648	d1_bun_min	1.138697	gcs_unable_apache_0.0	1.047007
6	ventilated_apache_0.0	1.874987	d1_temp_min	1.128697	leukemia_0.0	1.021712

Fig. 8: A table of the Variance Inflation Factor (VIF) scores of our features. This score tells us how easily each feature's value can be predicted from the other features. A score of 1.0 means no multicollinearity, scores <5 are not concerning, and scores >10 are of significant concern. The four blood pressure features have unusually high VIF scores (>75), and our corrective action here is to keep only "d1_mbp_min".

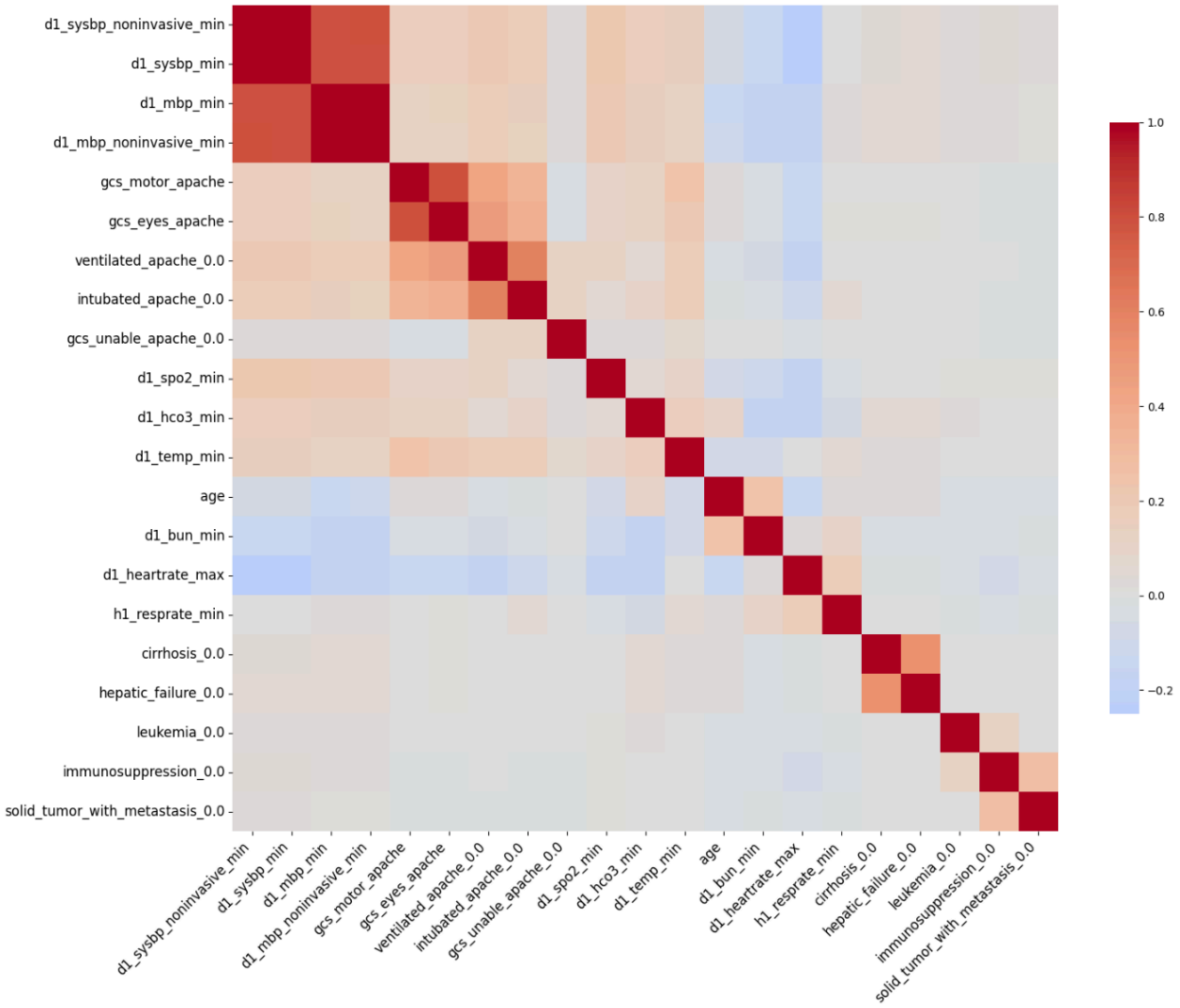


Fig 9: Despite our Lasso regularization step, we can see here that some sub-groups of features remained highly correlated with each other.